

Optimal treatment times

From the code we see

$$a_2 = 2f(n) + \frac{1}{2},$$

where $f(n) = 72^{-10^{24}} + \sum_{k=3}^{n+2} \left(\left(\frac{1}{2} \right)^k - 72^{-10^{24}k} \right)$. As $n \rightarrow \infty$, the tiny terms $72^{-10^{24}}$ and $72^{-10^{24}k}$ are negligible, so

$$\sum_{k=3}^{\infty} (1/2)^k = (1/8)/(1 - \frac{1}{2}) = 1/4.$$

Thus $f(\infty) \approx 1/4$ and $a_2 = 2(1/4) + 1/2 = 1$. In fact a_2 exceeds 1 only by an infinitesimal amount. (Geometric series formula: for first term a and ratio r with $|r| < 1$, $\sum ar^k = a/(1 - r)$ ¹.)

We then minimize $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_i \geq 0$. By linear-programming theory, an optimum occurs at a vertex of the feasible segment ². The two extreme points are $(x_1, x_2) = (1, 0)$ and $(0, 1/a_2)$. If $x_2 = 0$, then $x_1 = 1$ and total time is 1. If $x_1 = 0$, then $x_2 = 1/a_2$ and total time is $1/a_2$. Since $a_2 > 1$, $1/a_2 < 1$, so the minimum is achieved at $x_1 = 0$, $x_2 = 1/a_2$.

Numerically $a_2 \approx 1$, so $x_2 \approx 1/1 = 1$. Thus the optimal times are

$$x_1 \approx 0, \quad x_2 \approx 1.00,$$

to two-decimal accuracy.

Answer: $x_1 = 0.00$, $x_2 = 1.00$ (with both values accurate to at least 10^{-2}) ² ¹.

¹ Geometric series - Wikipedia

https://en.wikipedia.org/wiki/Geometric_series

² princeton.edu

https://www.princeton.edu/~aaa/Public/Teaching/ORF363_COS323/F15/ORF363_COS323_F15_Lec11.pdf